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RESEARCH ARTICLE

Learning Robust Perception-Based Controller for Quadruped Robot

FIKIH MUHAMAD[1], JUNG-SU KIM[1], (Member, IEEE), AND JAE-HAN PARK[2], (Member, IEEE)

[1]Research Center for Electrical and Information Technology, Department of Electrical and Information Engineering, Seoul National University of Science and Technology (SeoulTech), Seoul 01811, South Korea

[2]AI Robot Research and Development Department, Korea Institute of Industrial Technology (KITECH), Ansan 15588, South Korea

Corresponding author: Jung-Su Kim (jungsu@seoultech.ac.kr)

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ABSTRACT The perception of a quadruped robot is crucial in determining how the robot has to move while crossing challenging terrains. However, robustness in the quadruped perception is still a dilemma when facing a slippery, deformable, and reflective terrains. This problem arises due to the presence of noises from slippage on the foothold and misinterpretation of the terrain. The noises can elicit imperfect robot states and lead to the poor performance of the robot locomotion. This paper presents a robust perception-based control policy as a solution to overcome noises in the robot's perception. In particular, attention is paid to devise a robust perception method against noises both in proprioceptive and exteroceptive observations. In other words, the proposed control policy has a capability to estimate states and reduce effect of noises from both observations in the robot. A student-teacher algorithm is leveraged to train the control policy on a randomly generated terrain. Multiple robots also are employed in parallel learning regimes to reduce the training duration. The result is a robust control policy that can produce the optimal actions even when the robot's perception is affected by the noises observed not only the proprioceptive but also exteroceptive sensors. The robust control policy can handle a higher ratio of noise compared to the previous works in the literature. Robustness of the proposed control policy is validated using both the Isaac Gym simulation and a comparison with popular recurrent networks in the literature.

INDEX TERMS Gait controller, machine learning, quadruped robot, robust perception.

I. INTRODUCTION

Quadruped robot can navigate through many challenging terrains such as steps, slopes, and stairs, which are impossible to cross by a wheeled robot. That is possible because the quadruped robot has similar body properties to animals, which allows it to traverse terrains that animals can cross. However, the quadruped robot still cannot match the ability of animals while traversing challenging terrains.

The difference between the abilities of quadruped robots and animals lies mainly in the perception of terrains. The quadruped robot struggles to explicitly recognize the terrain. That happens because the robot's observations are affected by noises while traversing slippery, deformable, and reflective terrains. These noises can degrade the robot's performance. In contrast, most animals perceive the upcoming terrain using their eyes and sense the terrain with the tips of their feet. This foresight behavior enables the animal to recognize the properties of the terrain before getting a secure foothold. To attain capabilities on par with those of animals, the quadruped robot requires a robust control policy with high resistance to noises in its perception.

The quadruped robot utilizes proprioceptive observations from inner sensors such as an IMU (Inertial Measurement Unit) and joint encoders, to determine the body states of the robot. The locomotion controller uses these body states as feedback information to develop possible actions for the robot's movement. However, these proprioceptive

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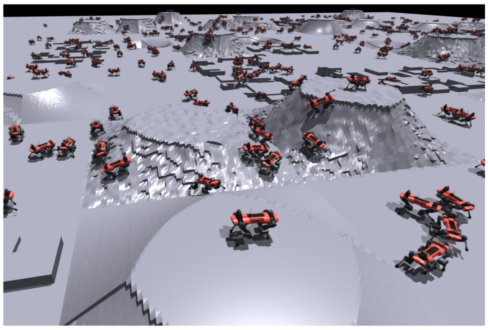

FIGURE 1. Multiple Anymal robot move in randomly generated terrain in Isaac gym.

observations often suffer from noises when the robot traverses slippery terrains. When the robot traverse the slippery terrain, the slippage on the foothold cannot be directly measured by using the inner sensors, which can cause the locomotion controller to produce inaccurate movements due to the incorrect information.

Another problem arises in exteroceptive observations from light-based sensors such as lidar or camera. The quadruped robot relies on these observations to perceive the terrain's information. This terrain information is crucial in getting a secure foothold before making a contact. However, exteroceptive observations are also easy to be contaminated by noises when the robot traverses on deformable and reflective terrains. General light-based sensors such as lidar and cameras are unable to notice reflective terrains due to their high reflectance. The light-based sensor is also incapable of distinguishing between unstable and rigid terrains. As a result, the robot may struggle to get a secure foothold, which leads to the ineffective robot movement.

In previous research, it was assumed that the observations are free from noises and the terrains are solid [1], [2], [3], [4], [5]. This assumption leads the locomotion controller to focus solely on producing the optimal locomotion while ignoring uncertainties in observations. This approach works well under ideal conditions but performs poorly in the presence of noises or terrains with uncertainties. Several approaches have aimed to estimate the robot slippage by modeling the terrain [6], [7]. However, these estimation methods rely on tuning terrain parameters and often suffer from insufficient data, especially when dealing with random factors such as deformable terrains. Other research has employed sensors in the robot's feet to estimate the slippage [8], [9]. Nonetheless, the use of foot sensors may not be practical when the robot needs to navigate in large environments, as these sensors can only measure a limited neighboring area.

Data-driven approaches, such as Reinforcement Learning (RL), have emerged as an alternative for addressing the issue of noisy observations in quadruped robots [10], [11], [12]. In RL, the control policy is updated by collecting data and calculating rewards based on current actions. In RL simulations, training data can be modified using domain randomization to simulate natural noises [13], [14], [15]. This optimal control policy can generate possible movements even in the presence of noises. However, this approach might show poor performance when the noise ratio in robot observations exceeds to some extent.

Recently, a robust perceptive locomotion approach has been developed by using sequential proprioceptive observations to estimate terrain properties in the absence of exteroceptive sensors [16]. This method allows the robot to produce foots reflexes even when its foot are trapped or slips during the locomotion. However, this approach can only perceive a small area nearby the robot. The result in [17] has employed a belief encoder to estimate the ground truth value of the terrain in noisy exteroceptive observations, but it focuses only on noises and biases in exteroceptive observations while assuming proprioceptive observations are perfect. Hence, this approach is inadequate for traversing the slippery terrain, especially when there are noises caused by the slippage.

Considering these exiting works, this paper presents a robust perception-based control policy to overcome effects of noises in robot observations. The contributions of this work are the following:

- Proposes a robust control policy that can produce the optimal locomotion even with noises in both exteroceptive and proprioceptive observations. The previous work only deals with noises in exteroceptive observations [17].
- Develops a control policy that has ability to estimate states and reduce noises in the robot's observations. The control policy can produce an appropriate movement even under a higher noise ratio than in [13].
- Evaluates the performance of the proposed approach through a comparative analysis with popular recurrent networks using simulation results. The previous approach only demonstrated the performances of the system in nature [16], [17], while this paper conducts a detailed analysis of several elements individually.

II. PRELIMINARIES

Sytem model: A quadruped robot is assumed as a floating base with four connected legs that can be moved by rotating the rotational joints. Define the robot state as $v_t^B, \omega_t^B \in \mathbb{R}^3$, the linear and angular velocities of the robot's base B with respect to the global initial frame respectively. $g_t^p = proj_R\, g$ represents the gravity projection concerning to the attitude of the robot where $g \in \mathbb{R}^3$ is the gravity vector in the global frame and $R \in \mathbb{R}^3$ is the base rotation vector. $q_t, \dot{q}_t \in \mathbb{R}^{12}$ are joints angular positions and velocities.

Command and action: The robot moves based on the high-level commands c_t as output references. $c_t = \left[v_x^r \ v_y^r \ \omega_z^r\right]$ represents tracking commands with v_x^r, v_y^r being body velocities target in x and y directions, and ω_z^r is the yaw rate target. To achieve corresponding legs' movements, the robot uses actions $a_t \in \mathbb{R}^{12}_{\text{clips}}$ as torque targets for each joints with clips to the maximum and minimum torque based on the robot's limitation.

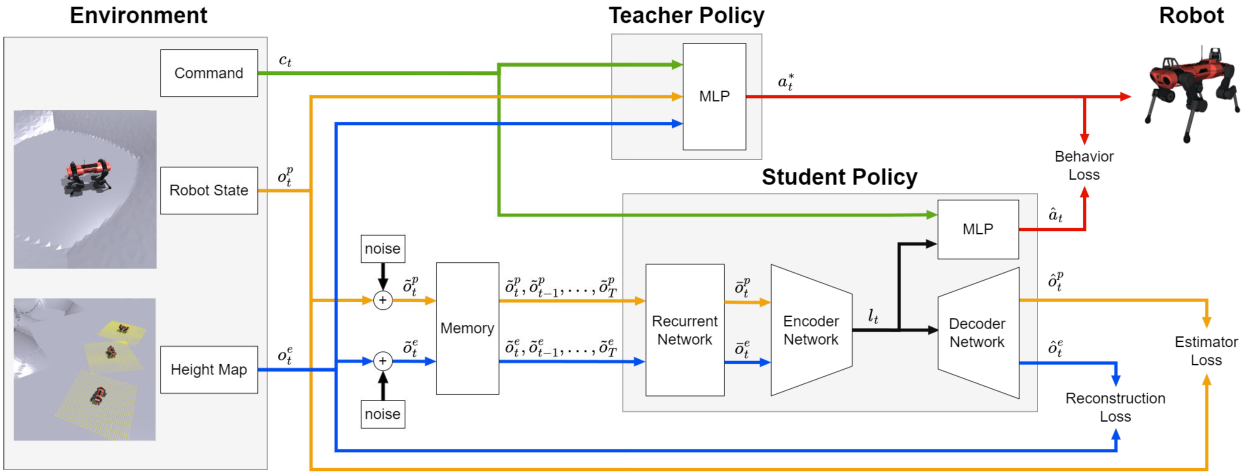

FIGURE 2. Illustration the overview of robust perception based controller training methods.

Proprioceptive observations: The robot needs information about its body state to determine the controller output. To simplify information from the system, proprioceptive observations o_t^p are defined as the measurement of internal sensors. This proprioceptive observations collects the necessary information about the robot's state. It can be expressed as:

$$o_t^p = \left[v_t^B \ \omega_t^B \ g_t^p \ q_t \ \dot{q}_t \ a_{t-1}\right] \in \mathbb{R}^{48} \tag{1}$$

Exteroceptive observation: Before making a contact, the robot has to perceive the terrain around it using a light-based sensor. The robot is assumed to be able to extract the robot-centring elevation map to recognize surrounding terrains. $o_t^e : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}^{34\times 20}$ represents the elevation map extraction from $-1.7\,m$ to $1.7\,m$ in the robot X-axis, and $-1\,m$ to $1\,m$ in the robot Y-axis with resolution $10\,cm$. The exteroceptive observations can be expressed as:

$$o_t^e = \begin{bmatrix} d_{(1.7,-1)} & d_{(1.7,0.9)} & \cdots & d_{(1.7,1)} \\ d_{(1.6,-1)} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ d_{(-1.7,-1)} & \cdots & \cdots & d_{(-1.7,1)} \end{bmatrix} \tag{2}$$

where $d_{(x,y)} = z^B - z_{(x,y)}$ serves as the distance measurement between the robot base height z^B and the terrain height $z_{(x,y)}$ in each coordinate around the robot's center $(x = 0, y = 0)$.

III. METHODOLOGY

A. OVERVIEW

This section presents the learning process for the robust control policy. The control policy aims to achieve robust locomotions despite noises in robot observations. It utilizes noisy observations from both exteroceptive and proprioceptive sensors to generate optimal actions in random terrains. In the training process, the teacher and student policies are introduced. The teacher policy represents the optimal control policy without any noises in observations, while the student policy corresponds to the proposed robust control policy that can operate properly even with noisy observations.

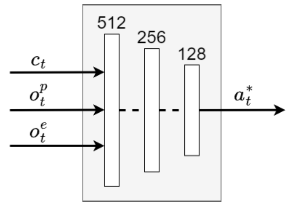

FIGURE 3. Teacher policy architecture.

The primary objective of the optimal teacher policy is to generate the best possible actions when traversing most areas of random terrains. To accomplish that, the teacher policy has access to noise-free observations as inputs while training in random terrains. This training phase aims to fully utilize the robot's capabilities, allowing the control policy to concentrate exclusively on generating the optimal control actions.

After obtaining the optimal teacher policy, the student policy is trained such that its output imitates the optimal actions generated by the optimal teacher policy even though only noisy observations are available. The student policy takes only noisy observations as inputs and predicts the optimal actions of the teacher policy as outputs. To predict the optimal actions, the student policy strives to mitigate the impact of noises in observations, enabling it to produce robust optimal actions. This is achieved by integrating the student policy with a combination of networks, including a recurrent, encoder, decoder, and MLP (Multi-Layer Perceptron) networks. The details of those networks in the student policy are described in the section III-D.

In the training process, the student-teacher algorithm is utilized to train the student policy to replicate the behavior of the teacher policy in the same terrain. Additionally, multiple Anymal models developed in [18] are employed to enhance the efficiency of the training. An overview of the learning process is illustrated in Figure 2.

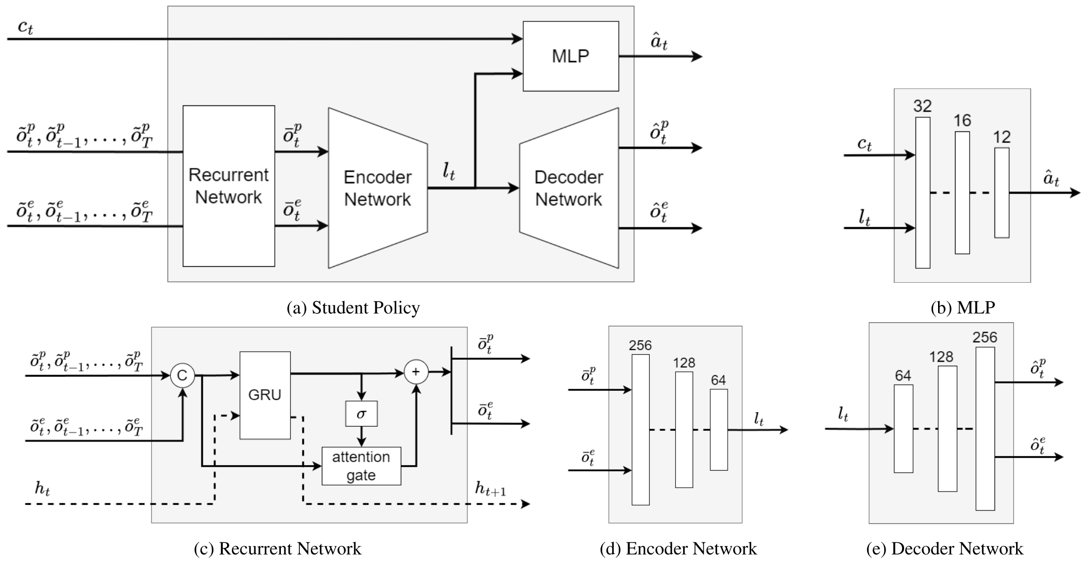

FIGURE 4. Student policy architecture.

B. TEACHER POLICY

The optimal teacher policy has to move in random terrains by generating the optimal actions. It is allowed to access noiseless exteroceptive (o^e_t) and proprioceptive (o^p_t) observations to maximize the control ability. Additionally, the teacher policy obtains commands c_t from the high-level navigation. These commands represent the base linear and angular velocity targets. To simplify information for the teacher policy, the teacher observations are defined as the construction of noiseless observations and commands. This teacher observations are represented as $o^{teacher}_t = (c_t, o^p_t, o^e_t)$.

The optimal teacher policy has to produce the optimal actions to make the robot move according to the velocity targets. To achieve this, Reinforcement Learning (RL) is used to train the teacher policy. The goal of reinforcement learning is to maximize the expected return that is the expectation of the sum of rewards over the given horizon. RL problem can be modeled as a Markov Decision Process (MDP), which is defined by $(s_t, a_t, r_t, a_{t+1}, s_{t+1})$. The MDP consists of a set of states $s \in \mathcal{S}$, a set of actions $a \in \mathcal{A}$, reward functions $r(s_t, a_t, s_{t+1})$, and a discount factor $\gamma \in [0, 1)$.

The teacher policy is treated as an agent in reinforcement learning. The output of the agent is a distribution, called a policy, over actions $a \in \mathcal{A}$ at states $s \in \mathcal{S}$. The agent's policy is denoted by $\pi(s_t|\theta)$ which is implemented by a multi-layer perceptron with parameter weights θ. The weights are updated by maximizing the expected returns $\mathrm{E}[\sum_{k=t}^{\infty} \gamma^k r_{t+k}]$ until getting the optimal policy $\pi(s_t|\theta^*)$.

The actions from the optimal teacher policy after the learning process can be represented as a Gaussian policy:

$$a^*_t \sim \mathcal{N}(\pi(o^{teacher}_t|\theta^*), \sigma I) \tag{3}$$

where σ is the variance in every actions. Figure 3 illustrates the teacher policy architecture.

C. NOISE IN OBSERVATION

The noises in observations can be presumed as the difference between sensor measurements results and ground truth values of the states. These noises can not be determined with exact values due to variations in the sensor accuracy. Moreover, the noises present in the observations are assumed to be Gaussian. Noises both in exteroceptive and proprioceptive observations can be expressed as:

$$\tilde{o}^p_t \sim \mathcal{N}(o^p_t, \sigma^p I) \tag{4}$$

$$\tilde{o}^e_t \sim \mathcal{N}(o^e_t, \sigma^e I) \tag{5}$$

where σ^p and σ^e represent variances of noises in proprioceptive and exteroceptive observations, respectively.

D. STUDENT POLICY

The student policy uses sequential noisy observations to predict the optimal actions in random terrains. Sequential noisy observations are constructed as a time series from time t until T. These sequential observations are expressed as:

$$X_t = \begin{bmatrix} \tilde{o}^p_t & \tilde{o}^p_{t-1} & \ldots & \tilde{o}^p_T \\ \tilde{o}^e_t & \tilde{o}^e_{t-1} & \ldots & \tilde{o}^e_T \end{bmatrix} \tag{6}$$

where the matrix X_t denotes sequential observations with noises, and T is the desired sequential time. To handle noises in observations, the student policy is comprised of a recurrent, encoder, decoder, and MLP networks. These networks serve to estimate the states and reduce the effect of noises in observations. Additionally, the student policy also gets commands c_t as body velocity targets. The commands and sequential noisy observations, altogether become information for student observations. The data used for the student policy is denoted by

$$o^{student}_t = [X_t, c_t] \tag{7}$$

The architecture of the student policy is inspired from [17], and illustrated in Figure 4.

1) RECURRENT NETWORK

At first, the sequential noisy observation X_t is injected into the recurrent network. If the noise ratio in the observations is excessively high, the noisy observations can result in unobservable states. In turn, it causes the state to become partially observed and leads to poor performance of the student policy. Therefore, the recurrent network can be used to prevent the states from becoming unobservable by estimating and making them observable. The recurrent network achieves this by extracting information from sequential observations, enabling the states to become observable. It works by determining how much the sequential observations have to pass through by learning gating factors to adapt to the sequential input. This learning process uses the hidden output h_t which contains the important information from the previous output as a feedback for the training. As the result, noisy sequential observations X_t are then extracted by the recurrent network into states estimate $(\bar{o}_t^p, \bar{o}_t^e)$. This recurrent network is illustrated in Figure 4c.

2) ENCODER NETWORK

The encoder network uses the states estimate $(\bar{o}_t^p, \bar{o}_t^e)$ from the recurrent network as the input. In this network, noises in the states estimate are reduced by squeezing the input to remove unimportant information such as noises. The network architecture can be designed by gradually decreasing the size of hidden layers from the input layer to the latent layer. The smallest size of the hidden layer produces the latent output l_t which contains the compressed information without unnecessary information like noises. The encoder network is illustrated in Figure 4d.

3) DECODER NETWORK

The latent output l_t from the encoder network is sent to the decoder network. In this network, the latent output is reconstructed to generate noiseless observations of the same size as their original values. The decoder is designed by progressively increasing the size of the hidden layers until the output layer matches the size of the encoder input layer. The output of the decoder consists of the predicted proprioceptive $\hat{o}_t^p$ and exteroceptive $\hat{o}_t^e$ observations. The decoder network is illustrated in Figure 4e.

4) MULTI-LAYER PERCEPTRON

The latent output l_t of the encoder is also transferred to the MLP (multi-layer perceptron) network in the student policy. The MLP predicts the optimal action $\hat{a}_t$ based on the latent output l_t and commands c_t from the high-level navigation. The MLP network is illustrated in Figure 4b.

In the student policy architecture, all of networks can be regarded as a unified deterministic policy $\mu(o_t^{student}|\phi)$ with shared parameter weights ϕ and ReLu activation function. The student policy is then trained using the student-teacher algorithm [19] to obtain the optimal deterministic policy. To be specific, the loss is defined as the difference the output (a_t^*) of the teacher policy and that $(\hat{a}_t)$ of the student policy for the purpose of making the student policy emulate the actions of the optimal teacher policy. The optimal student policy is represented as follows:

$$\hat{a}_t = \mu(o_t^{student}|\phi^*) \tag{8}$$

In this paper, the learning process is modified by incorporating noises in both exteroceptive and proprioceptive observations unlike the existing results in which noises are contained in only one of the two observations. Moreover, the noise ratio is also set higher than that described in the previous literature. This approach aims to enable the student policy to effectively handle any potential noises that may occur during locomotion in random terrains. However, the training time is a concern due to the computational costs involved. To address this, parallel learning [13] is employed, leveraging multiple robots to enhance the efficiency of the training process.

IV. RESULTS

A. EXPERIMENTAL SETUP

A custom-designed random terrain is utilized to evaluate this approach, as shown in Figure 1. The training process is conducted using the IsaacGym simulation on Ubuntu 18.04, employing multiple Anymal robots [18]. The terrain is intentionally designed to be close to the kinematic limits of the robot, encompassing flat planes, uneven steps, and mountain slopes. PyTorch is utilized to implement all of the algorithms, and the entire training process is executed on a single NVIDIA GeForce RTX 3090 GPU. Termination conditions are also taken into account, such as the robot entering a fall state, experiencing a body collision, or reaching the maximum step limit within an episode. In the event that any of these termination conditions are met, the robot is reset to its initial condition.

This paper performs only in the simulation to avoid the possible uncertainties in the real world. These uncertainties might occur due to nonlinearities in the robot and complexity in hardware and software.These uncertainties can afffect the dynamic characteristics of the robot and divert from the main objective. The simulation setup is designed by utilizing features such as friction, body inertia, sensor drifting with the purpose of having small reality gap by adding artificial noises. Several existing research in the literature also rely on similar simulation-based approaches to mitigate the impact of such uncertainties [10], [20], [21].

B. TRAINING TEACHER POLICY

Initially, the primary objective is to train an optimal teacher policy capable of traversing a wide range of random terrains without falling. The teacher policy relies on noiseless observations and randomly generated commands from the high-level navigation. Multiple Anymal robots are employed in the parallel training using the Proximal Policy Optimization (PPO) algorithm [13] to train the teacher policy. Parallel

TABLE 1. Noise ratio.

	Noise level	Noise ratio	Noise in [13]
Body linear velocity	$\pm 0.2\, m/s$	20%	$\pm 0.01\, m/s$
Body angular velocity	$\pm 0.4\, rad/s$	20%	$\pm 0.2\, rad/s$
Projected gravity	$\pm 0.15\, rad/s^2$	15%	$\pm 0.05\, rad/s^2$
Joint positions	$\pm 0.1\, rad$	10%	$\pm 0.01\, rad$
Joint velocities	$\pm 3.0\, rad/s$	15%	$\pm 1.5\, rad/s$
Height map	$\pm 0.2\, m$	20%	$\pm 0.1\, m$

TABLE 2. Hyperparameter.

Learning rate	0.0003
Sequences	5
Number of robot	500
Batch size	5000

training is utilized to address the sample inefficiency in reinforcement learning. By utilizing 5000 robots, the parallel training process is completed within a short duration of 30 minutes. The success of the teacher policy is determined by its ability to navigate the robot on the random terrain without experiencing falls, indicating its proficiency in generating optimal actions to tackle various terrain challenges. Once trained, the optimal teacher policy is utilized in subsequent stages without requiring any further fine-tuning.

C. ADDITIONAL NOISE

Once the optimal teacher policy that is capable of navigating most parts of the terrain is obtained, the training of the student policy begins under the conditions where the robot may encounter noises that affect its perception during the locomotion. These noises are assumed to follow a Gaussian distribution, as indicated in (4) and (5). The noises represent different failure cases that could potentially impact the measurement of proprioceptive and exteroceptive observations.

Initially, the optimal teacher policy freely explores the terrain using randomly generated commands. Subsequently, artificial noises are introduced to simulate potential observation errors that may occur in a real robot. The artificial noises are generated to mimic real-world phenomena during locomotion, such as sensor drifting, low friction in the foothold, stepping on deformable terrain, and perceiving a reflective surface.

These noises are directly injected into observations data with the noise ratio specified in Table 1. The resulting noisy observations are then stored in the memory and organized in a sequential manner.

D. TRAINING STUDENT POLICY

The student policy is expected to predict optimal actions $\hat{a}_t$ that closely resemble to the teacher's optimal actions by imitating the teacher's behavior. The student policy uses the same commands and terrain as the teacher policy. However, the student policy has access to only the noisy observations from the environment while the teacher policy can use the noiseless observations.

(a) Tracking error of the linear velocity of the body in x direction

(b) Tracking error of the linear velocity of the body in y direction

(c) Tracking error of the angular velocity of the body in yaw direction

FIGURE 5. (a) , (b), and (c) are the tracking errors from body velocities in the student policies A, B, C, and the proposed student policy with respect to target commands.

Estimator loss is defined as the error between proprioceptive predictions $\hat{o}_t^p$ and noiseless proprioceptive observations o_t^p. This estimation loss indicates how the student policy can estimate the ground truth robot states from noisy proprioceptive observations. Further, reconstruction loss is calculated as an error between exteroceptive predictions $\hat{o}_t^e$ and noiseless exteroceptive observations o_t^e. This reconstruction loss shows the error in the robot's perception of the original terrain from noisy exteroceptive observations. Additionally, the behavior loss is also calculated as the error between the optimal actions a_t^* of the teacher and the actions prediction $\hat{a}_t$ of the student. This behavior loss indicates the similarity between the student's and teacher's behavior.

In order to develop a student policy capable of effectively handling a higher ratio of noises compared to the approach presented in [13], the student-teacher algorithm [19] is employed. In this algorithm, the teacher policy generates the optimal actions using noiseless observations as input while

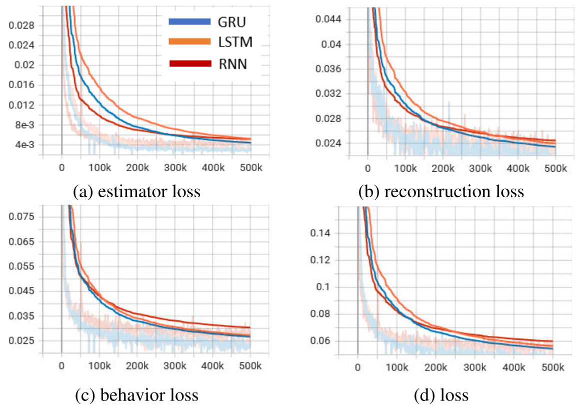

FIGURE 6. **Losses incurred during the training of student policies using popular recurrent networks: estimator loss (a) , reconstruction loss (b), behavior loss (c), and training loss (d).**

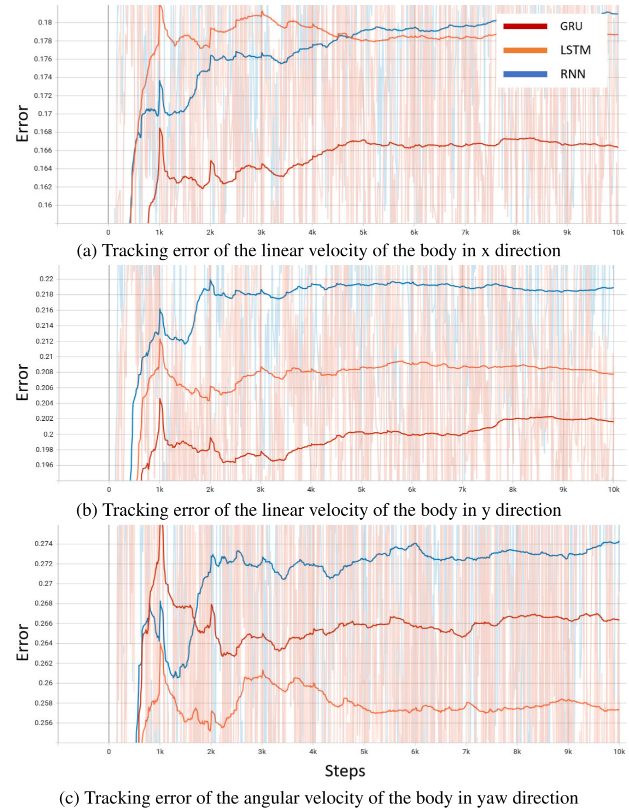

FIGURE 7. **(a) , (b), and (c) are the tracking errors in body velocities for each popular recurrent network within the student policies, with respect to the command reference.**

the student policy learns to emulate the teacher's behavior using latent information obtained from the encoder network. The student policy is trained using supervised learning where the noisy observations serve as the input and the corresponding noiseless observations act as the target output. During training, the student policy minimizes a combined loss function that includes the estimator, reconstruction, and behavior losses. The hyperparameters utilized in the training process are outlined in Table 2, and the loss for n robots is expressed as:

$$\mathcal{L}=\underbrace{\frac{1}{n}\sum_{i=1}^{n}(o_i^p-\hat{o}_i^p)^2}_{\text{estimator loss}}+\underbrace{\frac{1}{n}\sum_{i=1}^{n}(o_i^e-\hat{o}_i^e)^2}_{\text{reconstruction loss}}+\underbrace{\frac{1}{n}\sum_{i=1}^{n}(a_i^*-\hat{a}_i)^2}_{\text{behavior loss}} \quad (9)$$

The student policy can imitate the behavior of the teacher policy while traversing through random terrains by minimizing the loss function represented as (9). Minimizing the estimator loss enables the student policy to sense proprioceptive observations similar to the teacher policy. Additionally, minimizing the reconstruction loss allows the student policy to perceive terrain properties similar to the teacher policy. Subsequently, by minimizing the behavior loss, the student policy can accurately predict actions that closely resemble the optimal actions of the teacher policy.

E. VALIDATE THE ROBUSTNESS

Previous approaches in the literature frequently fail to consider the existence of noises in both exteroceptive and proprioceptive observations. Instead, the previous approaches tend to concentrate solely on noises present in the one type of observation. This limited perspective can lead to a lack of robustness in the performance when both types of observations are affected by noises. In contrast, this paper aims to tackle this issue by explicitly considering and addressing the presence of noises in both exteroceptive and proprioceptive observations.

To validate the proposed approach, four types of student policies are compared. First, the student policy A is trained without any noises. Second, the student policy B is trained using noises only in exteroceptive observations assuming perfect proprioceptive observations. Third, student policy C corresponds to the opposite of student policy B as it is trained using noises solely in proprioceptive observations while assuming perfect exteroceptive observations. The noise ratio for observations in student policy B and C are given in the Table 1. Last, the proposed student policy is trained using both noisy exteroceptive and proprioceptive observations employing the same noise ratio as described in Table 1.

After obtaining the trained student policies, their performance is evaluated by navigating in the random terrain using randomly generated commands. Both the proprioceptive and exteroceptive observations are added with noises according to the noise ratio specified in Table 1. The performance is quantified based on the minimum command tracking errors achieved by each policy. The smaller error indicates more robustness of the student policy.

Figure 5 presents the results, highlighting that the proposed student policy outperforms student policies A, B, and C with the smallest error. It proves that the proposed student policy has better performance than all other student policies. The proposed student policy excels in accurately tracking the target body linear velocities in the x and y directions, as well as the body yaw rate target. Contrarily, student policies A, B, and C struggle to fulfill the target commands, particularly in the case of yaw rate. These results validate that the proposed policy leads to robust perception against noises from both measurement sources.

The error is typically low due to the limited number of steps per episode during the early phase of the simulation. As the simulation progresses, the error values increse due to the dynamics of the system affected by noises. The efficiency of the method is also demonstrated through the use of parallel learning, resulting in 5 hours training time for each student policy. This represents a significant improvement compared to the existing methods in the literature.

F. POPULAR RECURRENT NETWORK

The recurrent network is applicable for predicting the sequence data. Several recurrent models, such as RNN (Recurrent Neural Network), LSTM (Long Short-Term Memory), and GRU (Gated Recurrent Unit), are popular in many research fields. However, choosing a suitable recurrent model for a specific task remains a question, especially when considering the LSTM and GRU networks.

In natural language processing, the GRU network has an advantage in handling a long text and small datasets compared to the LSTM network. Additionally, the structure of a GRU network is simpler, saving a significant amount of time compared to the LSTM. Regardless, a LSTM has an advantage over the GRU when dealing with complex data, as the GRU may suffer from a serious performance loss in such cases [22]. Drawing from this experience, this section aims to evaluate the performance of these popular recurrent networks in order to determine the most suitable one. The RNN, LSTM, and GRU networks are integrated into the recurrent network component of the student policy. All of these networks are trained using the same field and configuration. Once the student policies have been trained, they are executed with random target commands, and their performance is evaluated by calculating the error between the target commands and their body velocities.

Figure 6 shows that the GRU network outperforms both RNN and LSTM networks during the training process. Despite initial challenges in convergence, the GRU network successfully achieves the lowest loss at the end of training. Notably, the GRU network demonstrates better performance in predicting noiseless proprioceptive and exteroceptive observations, as evidenced in estimator and reconstruction losses. Although the LSTM network exhibits a behavior loss similar to that of the GRU network, the GRU network consistently maintains the lowest loss in this aspect as well.

During the validation test, the GRU network emerges as the best network with the lowest tracking error, as illustrated in Figure 7. The GRU network performs better in tracking body linear velocities target in the x and y directions. However, the LSTM network outperforms in terms of the yaw rate target. Those imply that while the GRU network stands out as the best recurrent network, the LSTM network still holds potential for utilization within this student policy.

V. CONCLUSION

This work shows the performance of an optimal student policy composed of various networks, enabling it to navigate random terrains even in the presence of noises both in exteroceptive and proprioceptive observations. The student policy consists of recurrent, encoder, decoder, and MLP networks. The recurrent network handles noisy observations and estimates the states to mitigate the presence of unobservable states. The encoder and decoder networks work together to reduce noises and reconstruct the ground truth values of the observations. Lastly, the MLP network utilizes latent information from the encoder network to predict optimal actions. This design has been proven to enhance the robustness of the control policy in the presence of noises in observations. Furthermore, the effectiveness of the parallel training has been demonstrated, resulting in shorter training times compared to previous approaches in the literature. Additionally, the performance of popular recurrent networks has been assessed, with the GRU network emerging as the most effective option.

In this paper, the terrains on the map are designed close to the robot's kinematic limits, with the assumption that the policy can traverse most areas without the need to avoid obstacles. Future works could introduce obstacles that the robot must navigate around, necessitating the optimization of obstacle avoidance strategies [23], [24], [25]. Furthermore, this study sets the hyperparameters only once, as it employs a single map in the simulation. Future research could utilize various maps to enhance the method's generalization. To achieve this, the utilization of adaptive hyperparameter tuning methods can improve the learning efficiency [26], [27].

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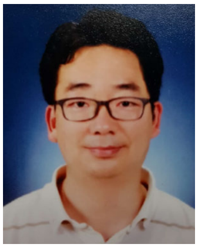
FIKIH MUHAMAD received the B.S. degree in electrical engineering from Universitas Indonesia, in 2020. He is currently pursuing the master's degree with the Department of Electrical and Information Engineering, Seoul National University of Science and Technology (SeoulTech). His research interests include robotics and artificial intelligence.



JUNG-SU KIM (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Korea University, in 1998, 2000, and 2005, respectively. From 2005 to 2008, he was a Postdoctoral Researcher with Seoul National University; the University of Stuttgart, Germany; and the University of Leicester, U.K. Since 2009, he has been with the Department of Electrical and Information Engineering, Seoul National University of Science and Technology (SeoulTech). His research interests include MPC, machine learning, and their applications.

JAE-HAN PARK (Member, IEEE) received the B.S. degree in electronic engineering from Dong-A University, in 1998, the M.S. degree from Pusan National University, in 2000, and the Ph.D. degree in electrical engineering from Korea University, in 2017. He is currently a Principal Research Scientist with the AI Robot Research and Development Department, Korea Institute of Industrial Technology (KITECH), Ansan, South Korea. His research interests include 3D perception for intelligent robots, robot motion planning, and AI-based robot control.